MATH441/MATH541
Applied Mathematics II: Complex Variables
Lecture slides 18: Residue Theorem

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March 8, 2023

Lecture outline

1. The Residue Theorem
2. Residue At A Pole Of Order m.
3. The argument principle theorem
4. Rouché Theorem
5. Fundamental theorem of algebra

The Residue Theorem. Part 1.

Definition

Let f have an isolated singularity at z_0. In some annulus $0 < |z - z_0| < R$, f has a Laurent expansion

$$f(z) = \sum_{n=-\infty}^{\infty} c_n (z - z_0)^n .$$

The number c_{-1} is called the *residue* of f at z_0, and is denoted by $\mathrm{Res}(f, z_0)$.

Thus the residue of f at z_0 is the coefficient of $1/(z - z_0)$ in the Laurent expansion of f. From Laurent's theorem

$$c_n = \frac{1}{2\pi i} \oint_\gamma \frac{f(z)}{(z - z_0)^{n+1}} dz$$

where γ is a positively oriented closed path about z_0 in the annulus $0 < |z - z_0| < R$.

The Residue Theorem. Part 2.

Hence

$$c_{-1} = Res\left(f, z_0\right) = \frac{1}{2\pi i} \oint_{\gamma} f\left(z\right) dz.$$

Therefore,

$$\oint_{\gamma} f\left(z\right) dz. = 2\pi i \, Res\left(f, z_0\right).$$

Hence, knowing the residue of f at z_0 is equivalent to evaluating the integral above.

The conclusion just stated is limited since we have required that γ enclose only one singularity of f. However, the residue theorem extends this result to the case where γ encloses any finite number of singularities of f.

The Residue Theorem. Part 3.

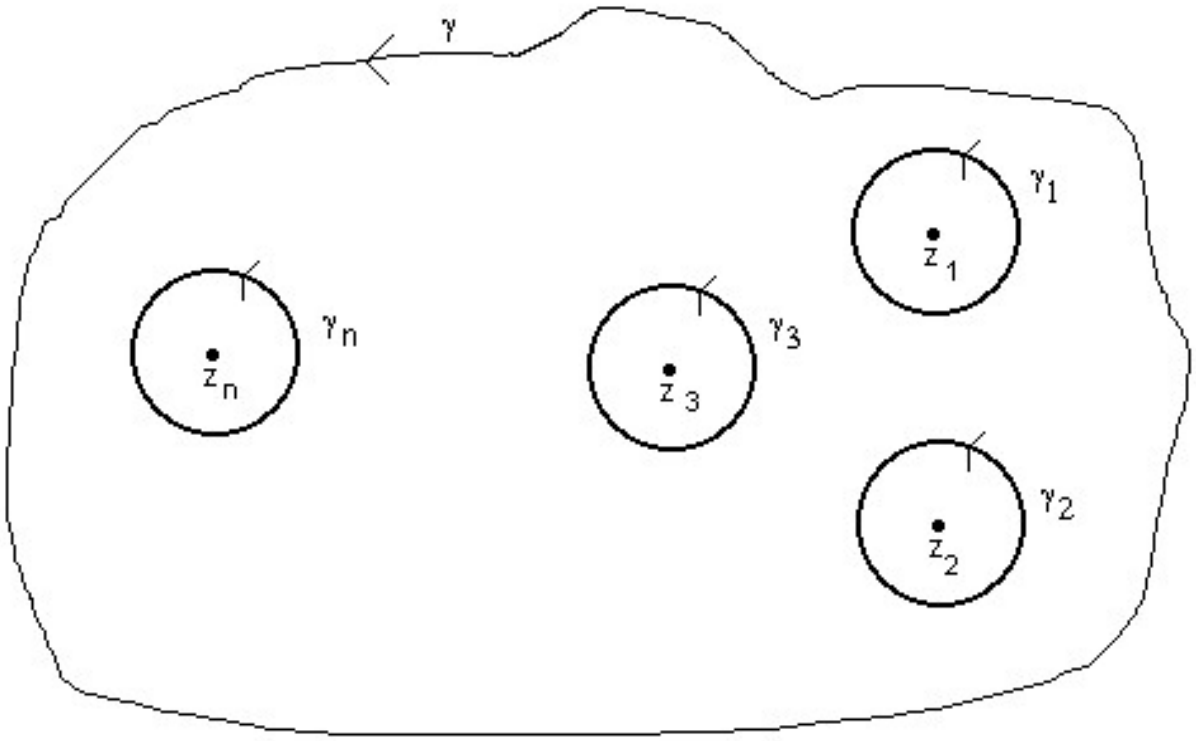
Theorem

Residue Theorem *Let γ be a positively oriented closed path, and let f be differentiable on γ and at all points interior to γ except at $z_1, z_2, \ldots, z_n$, which are isolated singularities of f. Then*

$$\oint_\gamma f(z)\,dz = 2\pi i \sum_{j=1}^{n} Res\,(f, z_j)$$

The Residue Theorem. Part 4.

Proof: Enclose each singularity z_j with a closed path γ_j about z_j, so that each γ_j is in the interior of γ and no two of $\gamma, \gamma_1, \ldots \gamma_n$ intersect, and each γ_j encloses just one singularity z_j of f.



The Residue Theorem. Part 5.

By the extended deformation theorem

$$\oint_{\gamma} f(z)\, dz = \sum_{j=1}^{n} \oint_{\gamma_j} f(z)\, dz$$

However,

$$\oint_{\gamma_j} f(z)\, dz = 2\pi i\, Res\,(f, z_j)$$

for $j = i, 2, \ldots, n$

Residue At A Pole Of Order m.. Part 1

Theorem

Residue At A Pole Of Order m.
Let f have a pole of order m at a. Then

$$Res\left(f,a\right)=\frac{1}{(m-1)!}\lim_{z\to a}\frac{d^{m-1}}{dz^{m-1}}\left(\left(z-a\right)^{m}f\left(z\right)\right).$$

Note: For $m=1$, $(m-1)!=1$ and the zeroth derivative is just the function itself.

Residue At A Pole Of Order m.. Part 2

Proof: Since f has a pole or order m at a, the Laurent expansion of f in some annulus about a is

$$f(z) = \frac{c_{-m}}{(z-a)^m} + \frac{c_{-m+1}}{(z-a)^{m-1}} + \cdots + \frac{c_{-1}}{(z-a)} + \sum_{n=0}^{\infty} c_n (z-a)^n$$

$$(z-a)^m f(z) = c_{-m} + c_{-m+1}(z-a) + \cdots + c_{-1}(z-a)^{m-1} + \sum_{n=0}^{\infty} c_n (z-a)^{n+m}$$

Differentiating m times we get

$$\frac{d^{m-1}}{dz^{m-1}}\left((z-a)^m f(z)\right) = (m-1)(m-2)\cdots 1 \cdot c_{-1} + \\ + m(m-1)\cdots 2 \cdot 1 \cdot c_0 (z-a) + \cdots$$

Thus $\quad \lim_{z\to a} \frac{d^{m-1}}{dz^{m-1}}\left((z-a)^m f(z)\right) = (m-1)!c_{-1} = (m-1)!\mathrm{Res}(f,a)$

Example 1

Find the residues of $f(z) = \dfrac{z^2 - 2z}{(z+1)^2 (z^2+4)}$.
This function has a double pole at $z = -1$ and simple poles at $z = \pm 2i$.
The residue at $z = -1$ is

$$\lim_{z \to -1} \frac{1}{1!} \frac{d}{dz} \left[(z+1)^2 \frac{z^2 - 2z}{(z+1)^2 (z^2+4)} \right]$$
$$= \lim_{z \to -1} \left[\frac{(z^2+4)(2z-2) - (z^2-2z)(2z)}{(z^2+4)^2} \right] = -\frac{14}{25}$$

The residue at $z = 2i$ is

$$\lim_{z \to 2i} \left[(z-2i) \frac{z^2 - 2z}{(z+1)^2 (z-2i)(z+2i)} \right] = \frac{-4-4i}{(2i+1)^2 (4i)} = \frac{7+i}{25}$$

Similarly, the residue at $z = -2i$ is $\frac{7-i}{25}$.

Example 2

Evaluate

$$\frac{1}{2\pi i}\oint_C \frac{e^{zt}}{z^2\left(z^2+2z+2\right)}dz.$$

The integrand has a double pole at $z=0$ and simple poles at $z=-1\pm i$. By straightforward calculations one can show that for $f(z)=\dfrac{e^{zt}}{z^2\left(z^2+2z+2\right)}$

$$\mathrm{Res}(f,0)=\frac{t-1}{2} \qquad \mathrm{Res}(f,-1+i)=\frac{e^{(-1+i)t}}{2\left(-1+i\right)^3}$$

$$\mathrm{Res}(f,-1-i)=\frac{e^{(-1-i)t}}{2\left(-1-i\right)^3}$$

Thus

$$\frac{1}{2\pi i}\oint_C \frac{e^{zt}}{z^2\left(z^2+2z+2\right)}dz=\frac{1}{2\pi i}\left(2\pi i\right)\times(\text{sum of the residues})$$

$$=\left[\frac{t-1}{2}+\frac{e^{(-1+i)t}}{2\left(-1+i\right)^3}+\frac{e^{(-1-i)t}}{2\left(-1-i\right)^3}\right]=\left[\frac{t-1}{2}+\frac{1}{4}e^{-t}\left(\sin t+\cos t\right)\right]$$

Applications: The argument principle theorem

Theorem

The Argument Principle. Let f be differentiable on a closed path Γ and at all points of its interior G, except at possibly a finite number of poles in G. Let Z be the number of zeroes of f in G and P the number of poles of f in G, with each zero and pole counted m times if its multiplicity is m. Then

$$\frac{1}{2\pi i}\oint_{\Gamma}\frac{f'(z)}{f(z)}dz = Z - P$$

The argument principle: Proof. Part 1

Proof: Note that the only singularities of f'/f that can occur in G must be at a zero of f or a pole of f.
Suppose that f has a zero of order k at z_0 in G. Then

$$f(z_0) = f'(z_0) = \cdots = f^{(k-1)}(z_0) = 0$$

whereas $f^{(k)}(z_0) \neq 0$. Hence in some open disk about z_0 the Taylor expansion of f about z_0 is

$$f(z) = \sum_{n=k}^{\infty} c_n (z - z_0)^n = \sum_{n=0}^{\infty} c_{n+k} (z - z_0)^{n+k}$$

$$= (z - z_0)^k \sum_{n=0}^{\infty} c_{n+k} (z - z_0)^n = (z - z_0)^k g(z)$$

where $g(z_0) \neq 0$ and $g(z)$ is differentiable at z_0. Hence

$$\frac{f'(z)}{f(z)} = \frac{k(z - z_0)^{k-1} g(z) + (z - z_0)^k g'(z)}{(z - z_0)^k g(z)} = \frac{k}{(z - z_0)} + \frac{g'(z)}{g(z)}.$$

The argument principle: Proof. Part 2

Since $g'(z)/g(z)$ is differentiable at z_0 it is clear from the last expression above that $f'(z)/f(z)$ has a simple pole at z_0 and

$$\operatorname{Res}\left(\frac{f'(z)}{f(z)}, z_0\right) = k.$$

Now suppose that f has a pole of order m at z_1 in G. Then the Laurent expansion of f in some annulus about z_1 is

$$f(z) = \sum_{n=-m}^{\infty} c_n (z - z_1)^n$$

Thus

$$(z - z_1)^m f(z) = \sum_{n=-m}^{\infty} c_n (z - z_1)^{n+m} = \sum_{n=0}^{\infty} c_{n-m} (z - z_1)^n = h(z)$$

The argument principle: Proof. Part 3

Now h is differentiable at z_1 and $h(z_1) = c_{-m} \neq 0$. Hence $f(z) = (z - z_1)^{-m} h(z)$, so that in some annulus about z_1

$$\frac{f'(z)}{f(z)} = \frac{-m(z - z_1)^{-m-1} h(z) + (z - z_1)^{-m} h'(z)}{(z - z_1)^{-m} h(z)} = \frac{-m}{(z - z_1)} + \frac{h'(z)}{h(z)}$$

Since $h'(z)/h(z)$ is differentiable at z_1 we see that $f'(z)/f(z)$ has a simple pole at z_1 and

$$\operatorname{Res}\left(\frac{f'(z)}{f(z)}, z_1\right) = -m$$

Thus the sum of the residues of $f'(z)/f(z)$ at its singularities in G counts each zero of f in G according to its multiplicity, and the negative of each pole according to its multiplicity.

The Argument Principle is key to the establishment of Rouché's Theorem.

Rouché Theorem.

Theorem

Theorem Rouché. If f and g are analytic inside and on a simple closed curve Γ and if

$$|f(z)| > |g(z)| \qquad \forall \ z \ \text{on } \Gamma,$$

then f and $f+g$ have the same number of zeros inside Γ, with the understanding that each zero is counted n times if its multiplicity is n.

Rouché Theorem. Proof. Part 1

Proof: Let $F(z) = g(z)/f(z)$ so that $g = fF$. Let N_1 and N_2 be the number of zeros inside Γ of $f+g$ and f respectively. Since f and g have no poles inside Γ, the Argument Principle says that

$$N_1 = \frac{1}{2\pi i}\oint_\Gamma \frac{f'+g'}{f+g}dz$$

and

$$N_2 = \frac{1}{2\pi i}\oint_\Gamma \frac{f'}{f}dz.$$

Thus

$$N_1 - N_2 = \frac{1}{2\pi i}\oint_\Gamma \frac{f'+f'F+fF'}{f+fF}dz - \frac{1}{2\pi i}\oint_\Gamma \frac{f'}{f}dz$$

$$= \frac{1}{2\pi i}\oint_\Gamma \frac{f'(1+F)+fF'}{f(1+F)}dz - \frac{1}{2\pi i}\oint_\Gamma \frac{f'}{f}dz$$

$$= \frac{1}{2\pi i}\oint_\Gamma \left[\frac{f'}{f} + \frac{F'}{1+F}\right]dz - \frac{1}{2\pi i}\oint_\Gamma \frac{f'}{f}dz = \frac{1}{2\pi i}\oint_\Gamma \frac{F'}{1+F}dz$$

Rouché Theorem. Part 2

However, $1 + F \neq 0$, because $F = -1$ implies that $|F| = 1$ or $|f(z)| = |g(z)|$, which is contrary to assumption. Hence $F'/(1+F)$ is analytic inside and on Γ. By the Cauchy integral theorem this means that the last integral on the right is 0 and therefore $N_1 = N_2$.

Rouché Theorem. Example

Example

How many zeros does the function

$$z^8 - 4z^5 + z^2 - 1$$

have inside the unit circle $|z| = 1$?

Let $f(z) = -4z^5$ and $g(z) = z^8 + z^2 - 1$. Then we may write the given polynomial as $f + g$. Also on $|z| = 1$ we have that

$$|f(z)| = \left|4z^5\right| = 4$$

and

$$|g(z)| = \left|z^8 + z^2 - 1\right| \leq \left|z^8\right| + \left|z^2\right| + 1 = 3.$$

Thus $|f| > |g|$ on $|z| = 1$. By Rouché's theorem the given polynomial has the same number of zeros as $f(z) = -4z^5$. But it is clear that $f(z)$ has a zero of order 5 at the origin and this is the only zero of f inside $|z| = 1$. Hence the given polynomial has 5 zeros inside the unit circle.

Fundamental theorem of algebra. Part 1.

Rouché's theorem leads to an elegant proof of the fundamental theorem of algebra.

Theorem

Fundamental theorem of algebra. Every polynomial of degree $n \geq 1$ has exactly n zeros.

Proof: Let $P(z) = a_0 + a_1 z + a_2 z^2 + \cdots + a_n z^n$, where $a_n \neq 0$. Let $f(z) = a_n z^n$ and $g(z) = a_0 + a_1 z + \cdots + a_{n-1} z^{n-1}$. Let C be a circle centered at the origin of radius $r > 1$. On C we have

$$\left|\frac{g(z)}{f(z)}\right| = \left|\frac{a_0 + a_1 z + a_2 z^2 + \cdots + a_{n-1} z^{n-1}}{a_n z^n}\right| \leq \frac{|a_0| + |a_1| r + |a_2| r^2 \cdots + |a_{n-1}| r^{n}}{|a_n| r^n}$$

$$\leq \frac{|a_0| r^{n-1} + |a_1| r^{n-1} + |a_2| r^{n-1} + \cdots + |a_{n-1}| r^{n-1}}{|a_n| r^n}$$

$$= \frac{|a_0| + |a_1| + |a_2| + \cdots + |a_{n-1}|}{|a_n| r}$$

Fundamental theorem of algebra. Part 2

$$= \frac{|a_0| + |a_1| + |a_2| + \cdots + |a_{n-1}|}{|a_n| r}$$

By choosing r large enough we can make $\left|\frac{f}{g}\right| < 1$, that is, $|f(z)| > |g(z)|$. The given polynomial $P(z) = f(z) + g(z)$ so by Rouché's Theorem $P(z)$ has the same number of zeros as $f(z) = a_n z^n$. But $a_n z^n$ has n zeros that are all located at $z = 0$.

Summation or Real Series

Let $P(z)$ be a polynomial of degree at least 2 with real coefficients and no integer zeros (although it may possess other real zeros). Let $z_1, \ldots, z_n$ be the zeros (real or complex) of $P(z)$. Then

$$\sum_{n=-\infty}^{\infty} \frac{1}{P(n)} = -\pi \sum_{j=1}^{m} Res\left(\frac{\cot(\pi z)}{P(z)}, z_j\right)$$

and

$$\sum_{n=-\infty}^{\infty} (-1)^n \frac{1}{P(n)} = -\pi \sum_{j=1}^{m} Res\left(\frac{\csc(\pi z)}{P(z)}, z_j\right).$$

We omit the proof and consider some examples.

Summation or Real Series. Example 1. Part 1

Example: Sum the series

$$\sum_{n=0}^{\infty} \frac{1}{n^2 + a^2},$$

where $a > 0$.

Let $P(z) = a^2 + z^2$. Then $P(z)$ is a polynomial of degree 2, with zeros at $z = \pm ai$. Also, let $f(z) = \frac{\cot(\pi z)}{P(z)}$ Thus

$$\sum_{n=-\infty}^{\infty} \frac{1}{n^2 + a^2} = -\pi \left[\operatorname{Res}(f, ai) + \operatorname{Res}(f, -ai)\right]$$

$$\operatorname{Res}(f, ai) = \frac{\cot(\pi ai)}{2ai} \quad \text{and} \quad \operatorname{Res}(f, -ai) = \frac{\cot(-\pi ai)}{-2ai} = \frac{\cot(\pi ai)}{2ai}.$$

Thus

$$\sum_{n=-\infty}^{\infty} \frac{1}{n^2 + a^2} = -\pi \left[\frac{2\cot(\pi ai)}{2ai}\right] = \frac{\pi i}{a} \cot(\pi ai)$$

$$= \frac{\pi i}{a} \frac{e^{-\pi a} + e^{\pi a}}{\left(\frac{1}{i}\right)\left(e^{-\pi a} - e^{\pi a}\right)} = \frac{\pi}{a} \left[\frac{e^{-\pi a} + e^{\pi a}}{e^{\pi a} - e^{-\pi a}}\right] = \frac{\pi}{a} \coth \pi a$$

Summation or Real Series. Example 1. Part 2

However, this is not the series we want to sum. Noting that $(-n)^2 = n^2$ we have

$$\sum_{n=-\infty}^{\infty} \frac{1}{n^2+a^2} = \sum_{n=-\infty}^{0} \frac{1}{n^2+a^2} - \frac{1}{a^2} + \sum_{n=0}^{\infty} \frac{1}{n^2+a^2}$$

$$= -\frac{1}{a^2} + 2\sum_{n=0}^{\infty} \frac{1}{n^2+a^2} = \frac{\pi}{a}\coth \pi a$$

Solving for the series that we want we get

$$\sum_{n=0}^{\infty} \frac{1}{n^2+a^2} = \frac{1}{2a^2} + \frac{\pi}{2a}\coth \pi a$$

For example, taking $a = 1$ we get

$$\sum_{n=0}^{\infty} \frac{1}{n^2+1} = 1 + \frac{1}{5} + \frac{1}{5} + \cdots = \frac{1}{2} + \frac{\pi}{2}\coth \pi$$

Summation or Real Series. Example 2

Show that

$$\sum_{n=-\infty}^{\infty} \frac{(-1)^n}{(n+a)^2} = \frac{\pi^2 \cos \pi a}{\sin^2 \pi a}$$

where a is real and different from $0, \pm 1, \pm 2, \ldots$

Let $P(z) = (z+a)^2$ and $f(z) = \dfrac{\csc \pi z}{P(z)}$. Then f has a double pole at $z = -a$ and

$$\mathrm{Res}(f, -a) = \lim_{z \to a} \frac{d}{dz} \left[(z+a)^2 \frac{\csc \pi z}{(z+a)^2} \right] = -\pi \csc \pi a \cot \pi a.$$

Hence

$$\sum_{n=-\infty}^{\infty} \frac{(-1)^n}{(n+a)^2} = \pi^2 \csc \pi a \cot \pi a = \frac{\pi^2 \cos \pi a}{\sin^2 \pi a}.$$

Thank you